

House Price Prediction – Project Summary

Objective

The objective of this project was to develop a machine learning model capable of predicting house prices based on various property characteristics such as area, number of bedrooms, bathrooms, stories, parking availability, furnishing status, and other amenities. The project aimed to identify the factors that have the greatest impact on property prices and compare the performance of different regression models.

Dataset Overview

The dataset consisted of 545 residential properties with 13 features describing different aspects of each house. The target variable was the house price. During the initial exploration, no missing values or duplicate records were found, making the dataset clean and suitable for analysis. Categorical features such as furnishing status, air conditioning, basement availability, and preferred area were converted into numerical format using one-hot encoding.

Data Analysis and Visualization

Several visualizations were created to better understand the data:

- A histogram was used to examine the distribution of house prices.
- A correlation heatmap was generated to identify relationships between house prices and property features.
- An Actual vs Predicted scatter plot was used to evaluate the prediction performance of the regression model.

The analysis showed that house prices are positively influenced by larger property area, more bathrooms, additional stories, parking availability, air conditioning, and preferred location.

Model Development and Evaluation

Two machine learning models were trained and evaluated:

1. Linear Regression
2. Random Forest Regressor

The dataset was split into 80% training data and 20% testing data. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Model Results

Model	MAE	RMSE	R ² Score
Linear Regression	970,043	1,324,507	0.653
Random Forest Regressor	1,013,969	1,398,116	0.613

Based on the evaluation metrics, Linear Regression performed slightly better than Random Forest Regressor and was selected as the best-performing model for this dataset.

Key Findings

- Property area was the most influential factor affecting house prices.
- Bathrooms, stories, parking spaces, and air conditioning significantly increased property value.
- Houses located in preferred areas generally had higher prices.
- Furnishing status also played an important role in determining market value.
- The house price distribution was right-skewed, indicating that only a small number of properties were extremely expensive.

Conclusion

This project successfully demonstrated the use of machine learning techniques for house price prediction. The Linear Regression model achieved the best overall performance with an R² Score of approximately 0.65, explaining around 65% of the variation in house prices. The results indicate that property size, amenities, and location are major factors influencing housing prices. These insights can help real estate businesses make better pricing decisions and understand which property features contribute most to market value.